

CMAN :=

Task 1 :=

$$PE = \left[\begin{array}{ll} PE[0][0] = 5 & PE[0][1] = 6 \\ PE[1][0] = 7 & PE[1][1] = 8 \end{array} \right]$$

Task 2 :=

Cycle 1 := We send in from A.

$$\begin{array}{l} \text{Row} \rightarrow 0 \\ \text{Row} \rightarrow 1 \end{array} A = \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right], \text{ which is the PE weights } \left[\begin{array}{ll} 5 & 6 \\ 7 & 8 \end{array} \right]$$

$$PE[0][0] = 0 + (1 \times 5) = 0 + 5 = 5$$

$$PE[0][1] = 0 + (1 \times 6) = 0 + 6 = 6$$

$$PE[1][0] = 0 + (3 \times 7) = 0 + 21 = 21$$

$$PE[1][1] = 0 + (3 \times 8) = 0 + 24 = 24$$

$$C_1 = \left[\begin{array}{ll} 5 & 6 \\ 21 & 24 \end{array} \right]$$

Cycle 2 := Now, Row 0 sends 2 and Row 1 sends 4.

$$PE[0][0] = 5 + (2 \times 5) = 5 + 10 = 15$$

$$PE[0][1] = 6 + (2 \times 6) = 6 + 12 = 18$$

$$PE[1][0] = 21 + (4 \times 7) = 21 + 28 = 49$$

$$PE[1][1] = 24 + (4 \times 8) = 24 + 32 = 56$$

$$C_2 = \left[\begin{array}{ll} 15 & 18 \\ 49 & 56 \end{array} \right]$$

$C_3 = \text{row } 0 \rightarrow 1, \text{ row } 1 \rightarrow 3$

$$PE[0][0] = 0 + (1 \times 5) = 0 + 5 = 5$$

$$PE[0][1] = 0 + (1 \times 6) = 0 + 6 = 6$$

$$PE[1][0] = 0 + (3 \times 5) = 0 + 15 = 15$$

$$PE[1][1] = 0 + (3 \times 6) = 0 + 18 = 18$$

$$C_3 = \begin{bmatrix} 5 & 6 \\ 15 & 18 \end{bmatrix}$$

$C_4 = \text{row } 0 \rightarrow 2, \text{ row } 1 \rightarrow 4$

$$PE[0][0] = 5 + (2 \times 7) = 5 + 14 = 19$$

$$PE[0][1] = 6 + (2 \times 8) = 6 + 16 = 22$$

$$PE[1][0] = 15 + (4 \times 7) = 15 + 28 = 43$$

$$PE[1][1] = 18 + (4 \times 8) = 18 + 32 = 50$$

$$\text{Final result } C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

TASK 3 :-

a) 8 total mac operations performed.

b) Every number was used 2 times.

c) There was one access for each value in A which is give 4 access, one access for each weight in B which gives another 4 accesses and one access for each answer written into C. Which gives the last 4 accesses. There are 12 memory accesses in total.

TASK 4: =

In weight stationary the weights of B are preloaded into the PEs and stay fixed throughout the computation. In output stationary, the partial sums of C stay fixed inside the PEs and accumulate. ~~Both~~ while both A and B values stream in from memory every cycle.